Reed Williams

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Education

Appalachian State University **Boone, NC**
B.S. Computer Science *Aug 2021 – May 2025*
Cumulative GPA: 3.68/4.0

Skills

- Built **TypeScript/Node.js** backends (**Express**, **REST APIs**) for real-time web applications.
- Designed data schemas and **ETL pipelines** to clean, correlate, and store multi-source scraped data at scale.
- Developed real-time systems (**Socket.io**) and interactive web apps (**HTML/CSS/JS**) with collaborative features.
- Open source/**Linux distribution packaging** contributions (**Nix/NixOS**).
- Automated deployments via **GitLab CI/CD** with **PyTest**, secret management, and containerization (**Docker**).
- Built RESTful API integrations for **SwiftUI** iOS app.
- Wrote **Python** automation scripts for testing and legacy code migration.
- Applied **OOP principles (Java)** and **relational database design (SQL)** for data architecture.
- Prototyped language concepts in **Haskell** and **Assembly** for academic exploration.
- Managed **Linux/macOS/Windows** development environments, including system configuration and shell scripting.

Projects

Not Drawful — Web-based Drawing Party Game
- Collaborated with frontend designer to implement game logic through **WebSockets**.
- Implemented backend functionality with **Node.js**, **Express**, and **Socket.io** for backend and game coordination.
- Developed and integrated real-time drawing canvas, allowing drawings to be shared for game functionality.

Hunger and Health Coalition — iOS App
- Collaborated with Hunger and Health Coalition (HHC), a nonprofit in Boone, NC, to develop an **iOS app**.
- Contributed to the app's functionality using **SwiftUI** and **Alamofire** to retrieve backend information.
- Coordinated with fellow students in the backend team to connect our work together.

Experience

Decidio.ai **Remote**
Data Engineer, Intern *June 2025 – Oct 2025*
- Designed and implemented a data model to support **Decidio**, optimizing for performance, flexibility, and growth.
- Developed **ETL** processes to clean, transform, and load data into structured storage (**MongoDB**).
- Implemented **LLM-driven data extraction pipeline** to quickly and accurately transform non-structured **HTML**.
- Built **TypeScript** scrapers with **Playwright** to extract data from dynamic sites (**SPAs, e-commerce**).
- Mapped raw scraped data to domain-specific types before **MongoDB** insertion with functionality to correlate data between different sites.

Billtrust **Remote**
Process Automation Engineer, Intern *May 2023 – Aug 2023*
- Worked at **Billtrust**, a billing services company specializing in invoice and remittance processing automation.
- Migrated codebases into the company's **GitLab** instance.
- Designed and implemented a **CI/CD pipeline**, incorporating secure secret management practices and automated testing using **PyTest**.